BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 148 · Lesson 148 · Network Hardware: Switches, Routers and Access Points

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a firewall generally use to permit or block a connection?

Q2. What data structure follows FIFO?

Q3. What is the result of an AND gate with inputs 1 and 0?

Q4. What is a foreign key used for?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

switch — Connects local devices and forwards data-link frames.

router — Forwards network-layer IP packets between networks.

access point — Connects wireless devices to a LAN.

NIC — Network interface controller for network connectivity.

New learning

Local connections — A switch forwards Ethernet frames using learned MAC addresses.

Across networks — Router consults IP destinations and routing information.

Wireless access — Access point bridges wireless stations onto a network.

Worked example

Two school laptops on one wired LAN share a file; one then accesses a website on another network. Trace the hardware.

1. Laptop and switch — Identify the facts and make an initial prediction.

2. Router/default gateway — Explain the mechanism or trace the intermediate stage.

3. Remote network — Within the LAN, NICs form frames and the switch forwards using MAC addresses. To reach another IP network, traffic is sent to a router/default gateway that forwards packets. The gateway rewrites the link framing per next hop.

We Do · Practise with guidance

A hall has strong wired connectivity but poor Wi-Fi. What hardware should be added, and why not another router by default?

Wired LAN works

Identify the facts and make an initial prediction.

Install suitable Wi-Fi AP

Explain the mechanism or trace the intermediate stage.

Clients join same LAN

Use a correctly connected wireless access point to extend radio coverage onto the existing LAN. An extra NAT router may create an unwanted separate network; plan channels, capacity and placement.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Differentiate switch, router and access point with their forwarding roles. [6]

Question 2

For a PC accessing a website, explain why both MAC and IP addresses are useful. [6]

Question 3

Design a network for two labs and a library with wireless devices and an internet connection. Label device placement and traffic path. [8]

Exit Ticket · Answer before checking

A basic Ethernet switch chooses internet routes using destination website names.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A policy of rules, for example addresses, ports or connection state.

Q2. A queue.

Q3. 0.

Q4. Referencing a candidate key in another relation.

You Do pages · Independent questions

Q1. Switch forwards LAN frames via MAC [2], router forwards IP packets among networks [2], AP connects wireless clients onto LAN [2].

Q2. IP identifies destination and enables routing [3]; MAC delivers frames on local link/next hop [3].

Q3. Switches/local ports [2], suitable access points [2], router/gateway [2], justified end-to-end path and coverage [2].

Exit Ticket · Model answer

False. Switches forward local frames; routers forward between IP networks.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: